Flax and Breast Cancer: A Systematic Review.

Background. Flax is a food and dietary supplement commonly used for menopausal symptoms. Flax is known for its lignan, α-linolenic acid, and fiber content, components that may possess phytogestrogenic, anti-inflammatory, and hormone modulating effects, respectively. We conducted a systematic review of flax for efficacy in improving menopausal symptoms in women living with breast cancer and for potential impact on risk of breast cancer incidence or recurrence. Methods. We searched MEDLINE, Embase, the Cochrane Library, and AMED from inception to January 2013 for human interventional or observational data pertaining to flax and breast cancer. Results. Of 1892 records, we included a total of 10 studies: 2 randomized controlled trials, 2 uncontrolled trials, 1 biomarker study, and 5 observational studies. Nonsignificant (NS) decreases in hot flash symptomatology were seen with flax ingestion (7.5 g/d). Flax (25 g/d) increased tumor apoptotic index ($P < .05$) and decreased HER2 expression ($P < .05$) and cell proliferation (Ki-67 index; NS) among newly diagnosed breast cancer patients when compared with placebo. Uncontrolled and biomarker studies suggest beneficial effects on hot flashes, cell proliferation, atypical cytomorphology, and mammographic density, as well as possible anti-angiogenic activity at doses of 25 g ground flax or 50 mg secoisolariciresinol diglycoside daily. Observational data suggests associations between flax and decreased risk of primary breast cancer (adjusted odds ratio [AOR] = 0.82; 95% confidence interval [CI] = 0.69-0.97), better mental health (AOR = 1.76; 95% CI = 1.05-2.94), and lower mortality (multivariate hazard ratio = 0.69; 95% CI = 0.50-0.95) among breast cancer patients. Conclusions. Current evidence suggests that flax may be associated with decreased risk of breast cancer. Flax demonstrates antiproliferative effects in breast tissue of women at risk of breast cancer and may protect against primary breast cancer. Mortality risk may also be reduced among those living with breast cancer.